PROJECT PRESENTATION

Ticket Resolution Multi-Agent-System

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AGENDA

1. The business, the data, the problem
2. Solution
3. Data
4. Request Flow
5. Multi-Agent Architecture
6. Trusting the AI
7. My approach
8. Challenge
9. Decisions
10. Questions

THE BUSINESS AND THE PROBLEM

- LUMA Wellness provides services in the wellness and beauty industry.
- Its customer support platform manages complaints related to appointments, payments, memberships, and bookings.
- Each complaint traditionally requires manual investigation by support staff across multiple systems.

PROPOSED SOLUTION

- Build an AI-assisted system that can investigate customer complaints and recommend appropriate resolutions.
- Use a multi-agent workflow to manage the process from complaint intake through investigation, policy review, recommendation, and escalation.
- Automatically resolve routine, low-risk cases when sufficient evidence is available.
- Require human review and approval for irreversible or financial actions, such as issuing refunds.
- Provide support staff with research tools to inspect evidence, review policies, and investigate cases directly within the platform.

DATA

We would be primarily dealing with two kinds of data

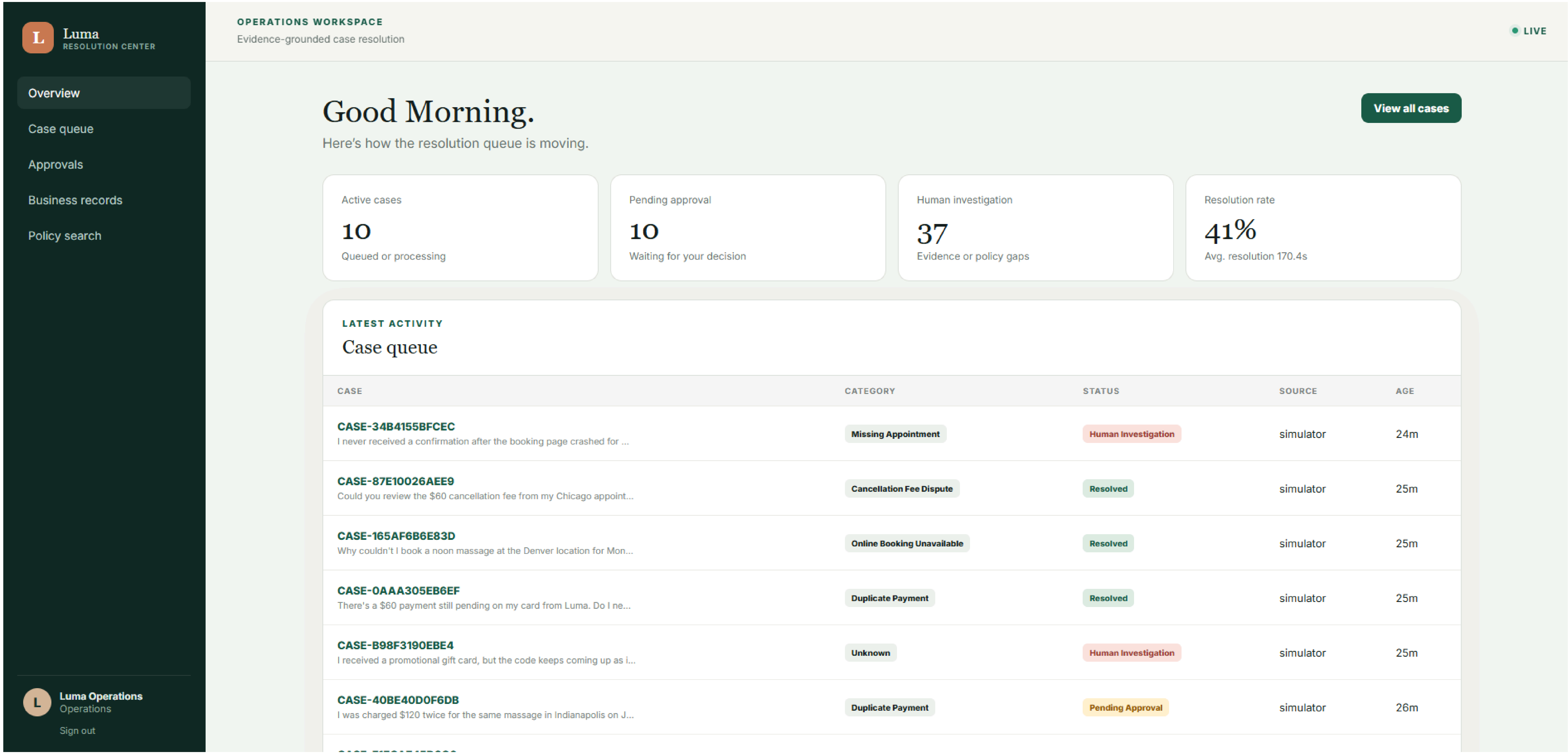
Operations Data

- Structured business records used by the agents to investigate customer complaints and establish evidence.
- Details like Customer info, memberships, appointments etc
- Stored in a persistent SQL/NoSQL database

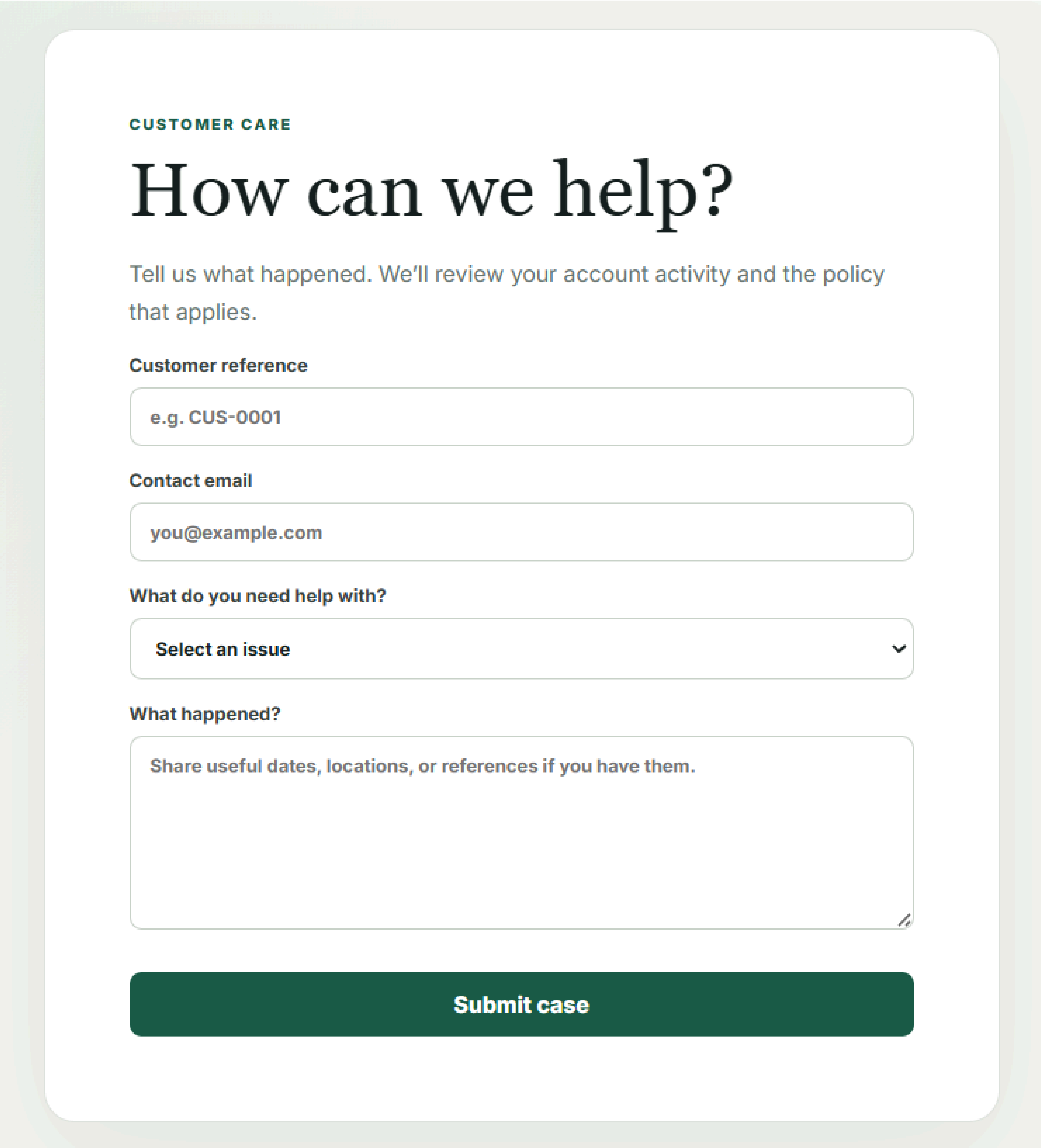
Policy Data

- Business rules used to determine the appropriate resolution for each complaint.
- Policies are temporal: the system retrieves the version that was effective when the incident occurred, rather than automatically using the latest policy.

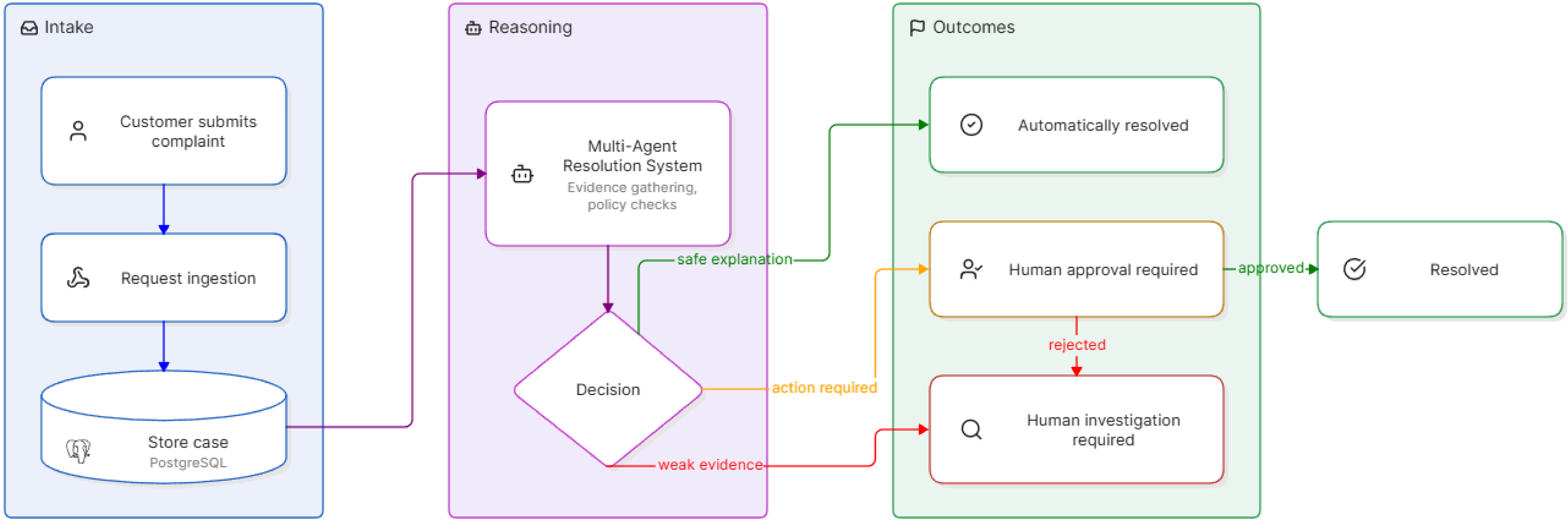
DASHBOARD OVERVIEW



CUSTOMER REQUEST INTAKE

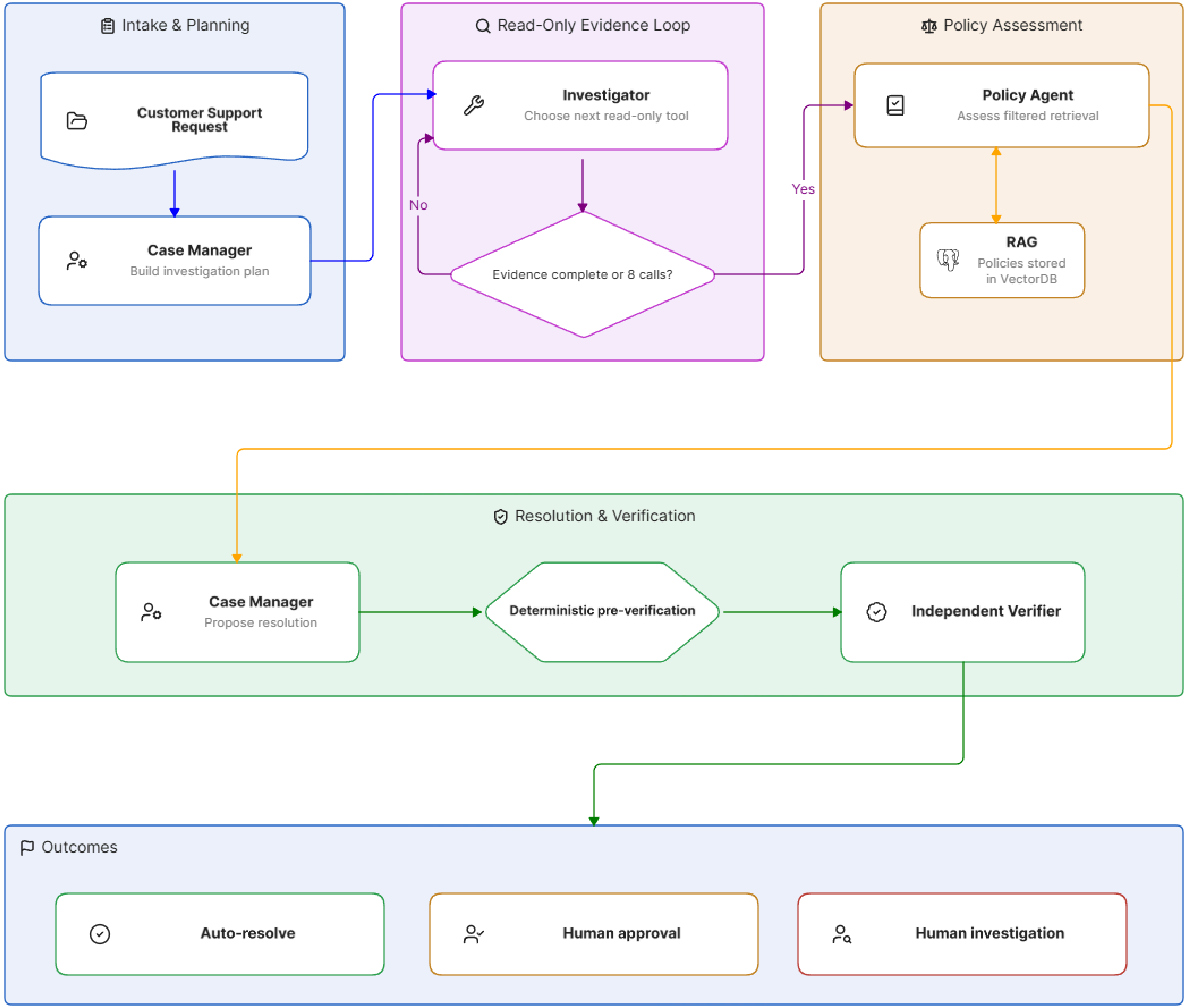


REQUEST FLOW



THE AI ARCHITECTURE

- 5 AI Agents
- 3 types of outputs



COMMS/HANDOFF - GRAPH STATE

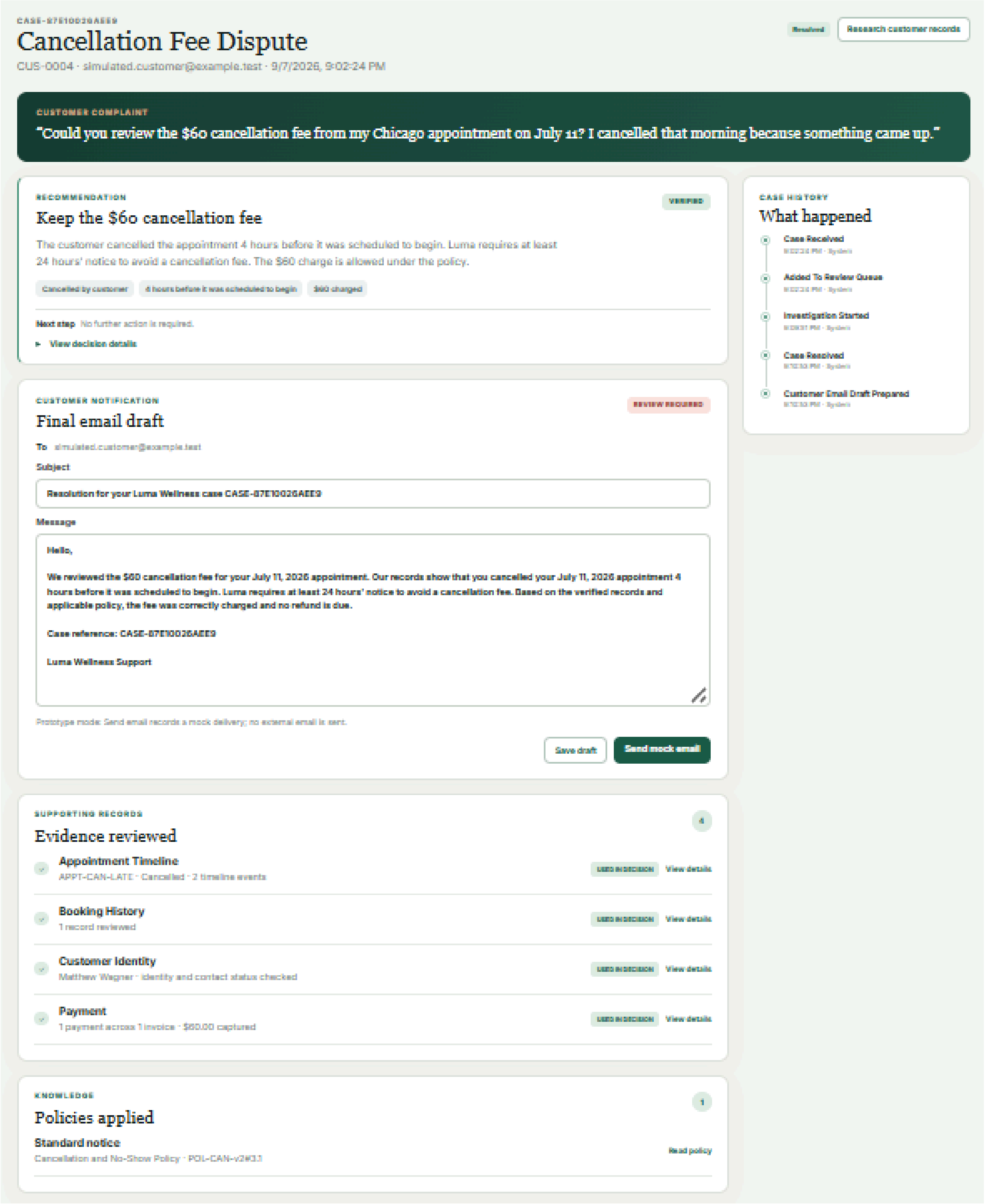
JSON


```json
{
  "case_id": "case-uuid",
  "case_reference": "CASE-8A42F190",
  "case_run_id": "71eeb913-1471-40ac-9ca5-a9f45f55f012",
  "case_received_at": "2026-09-08T10:15:00Z",
  "complaint_text": "The provider cancelled my appointment, but I was still charged an $80 cancellation fee.",
  "customer_reference": "CUS-0142",
  "claimed_category": "cancellation_fee_dispute",
  "evidence": [],
  "escalation_reason": null
}
```


Fields not populated yet:

plan
evidence_coverage
policy_results
policy_assessment
proposal
verification
final_disposition
approval_decision
execution_receipt
customer_response

EXAMPLE



Cancellation Fee Dispute
CUS-0004 · simulated.customer@example.test · 9/7/2026, 9:02:24 PM
Research customer records
CUSTOMER COMPLAINT
"Could you review the $60 cancellation fee from my Chicago appointment on July 11? I cancelled that morning because something came up."
RECOMMENDATION
VERIFIED
Keep the $60 cancellation fee
The customer cancelled the appointment 4 hours before it was scheduled to begin. Luma requires at least 24 hours' notice to avoid a cancellation fee. The $60 charge is allowed under the policy.
Cancelled by customer
4 hours before it was scheduled to begin
$60 charged
Next step No further action is required.
View decision details
CASE HISTORY
What happened
Case Received
Added To Review Queue
Investigation Started
Case Resolved
Customer Email Draft Prepared
CUSTOMER NOTIFICATION
REVIEW REQUIRED
Final email draft
To simulated.customer@example.test
Subject
Resolution for your Luma Wellness case CASE-87E10026AEE9
Message
Hello,
We reviewed the $60 cancellation fee for your July 11, 2026 appointment. Our records show that you cancelled your July 11, 2026 appointment 4 hours before it was scheduled to begin. Luma requires at least 24 hours' notice to avoid a cancellation fee. Based on the verified records and applicable policy, the fee was correctly charged and no refund is due.
Case reference: CASE-87E10026AEE9
Luma Wellness Support
Prototype mode: Send email records a mock delivery; no external email is sent.
Save draft
Send mock email
SUPPORTING RECORDS
Evidence reviewed
4
Appointment Timeline
APPT-CAN-LATE · Cancelled · 2 timeline events
View details
Booking History
1 record reviewed
View details
Customer Identity
Matthew Wagner · identity and contact status checked
View details
Payment
1 payment across 1 invoice · $60.00 captured
View details
KNOWLEDGE
Policies applied
1
Standard notice
Cancellation and No-Show Policy · POL-CAN-v2#3.1
Read policy

TRUSTING THE AI

To make the system safe, reliable, and trustworthy, I focused on two key areas:

1. Guardrails - Controlling what the system is allowed to do and when human approval is required.

2. Evaluations - Measuring whether the system makes accurate, evidence-backed, and safe decisions.

GUARDRAILS

- Structured Model Outputs - validated through pydantic schema checks
- Deterministic Pre-verification - checks for evidence and policy citations
- A seperate Verification agent - acting as llm as a judge
- Human approval gate - the agents dont have access to write tools.

EVALS

Golden Dataset

- 12 ground-truth business scenarios
- 60 behavioral cases: 36 development + 24 held-out
- Five test types: happy path, boundary, counterfactual, adversarial, and tool failure
- Evaluation truth is isolated from runtime agents

What each case specifies

- Expected classification, outcome, and disposition
- Required evidence and source records
- Required policy sections and tools
- Exact action type, target, and amount where applicable
- Forbidden actions and expected escalation reasons

Deterministic grading

- Decision, disposition, evidence, source, policy, and tool recall
- Exact action arguments
- Unauthorized and duplicate execution
- Latency, token usage, and estimated cost

So we could set thresold depending on the safety and risk nature of the problem.

RESULTS

Quality • RAG recall@5: 94.12% — 16/17 • Invalid temporal or scope results: 0 • Latest 10-case development slice: • Classification and decision accuracy: 100% • Source and tool recall: 100% • Disposition, evidence, policy, and action accuracy: 90% All-metrics case pass rate: 50% — 5/10	Safety • Unauthorized action rate: 0% • Duplicate execution rate: 0% • Model has no write tools • Financial actions stop for human approval • Missing or unsupported evidence fails closed
Latency • RAG mean: 11 ms • RAG P95: 26 ms • End-to-end P50: 59.2 seconds • End-to-end P95: 315.2 seconds	Cost (per request/complain) • Input: 43,644 tokens • Output: 2,104 tokens • Assumed cost: 0.0406/request (43,644 x 0.75/1M + 2,104 × \$3.75 / 1M)

MY APPROACH

- Understand the workflow - Map the complaint-resolution process, decisions, data, and policies.
- Decompose it into specialized agents - Give each agent one responsibility within an end-to-end workflow.
- Define clear contracts and shared state - Use structured inputs and outputs to make agent handoffs predictable.
- Make critical decisions deterministic - Use code-based checks, controlled tools, and approval gates around the LLM.
- Build for trust and continuous improvement - Add observability, evaluations, and feedback loops from the beginning.

CHALLENGE

1. Problem

- Too many cases ended in human investigation
- Even a single intermediate failure could escalate the entire case

2. What tracing revealed

- Model API timeouts and rate limits caused incomplete agent outputs
- Some deterministic checks were overly strict
- Technical failures and genuine evidence gaps followed the same escalation path

3. How I addressed it

- Added retries for transient model/API failures
- Introduced a backup model/provider
- Refined deterministic checks to reduce false escalations
- Preserved strict approval gates for irreversible actions

DECISIONS

1. Bounded autonomy
- Customer emails and all monetary actions require human review and approval.
2. Customer follow-ups deferred
- The current system works with available case data; multi-turn customer interaction is a future enhancement.
3. Evidence over assumptions
- Cases with missing or conflicting evidence are escalated instead of being resolved speculatively.
4. Autonomy can increase gradually
- Low-risk actions may be automated in the future after sufficient evaluation and production validation.
5. Batch API - could cost less

QUESTIONS